AML Single-Cell Causal Discovery & Drug Repurposing Dashboard

User & Exploration Guide: Interactive Graphical Modeling, In Silico Perturbations, Leukemic Hierarchy Rewiring & Clinical Pharmacology

Target Audience: Computational Biologists, Hematologist-Oncologists, Pharmacologists

Clinical Dataset: van Galen et al. (GSE116256 / AML1012-D0)

Inference Engine: Depth-Bounded PC Algorithm (depth=2, Fisher-Z)

Live Cloud Demo: aml-causal-discovery-7777.streamlit.app

1. Executive Overview & Rapid Deployment

Acute Myeloid Leukemia (AML) is characterized by blockades in hematopoietic differentiation and the clonal proliferation of immature leukemic blasts. Standard transcriptomic co-expression analyses (e.g., WGCNA) identify thousands of correlated gene pairs but fundamentally suffer from confounding: they cannot distinguish causative oncogenic drivers from passenger correlations, nor can they determine edge directionality ($X \to Y$ vs. $Y \to X$). This dashboard provides an autonomous, clinical-grade interface to explore a **rigorously constrained causal Directed Acyclic Graph (DAG)** derived from van Galen single-cell bone marrow aspirates.

System Prerequisites & Local Execution:

The application is engineered on top of Python 3.10+ and Streamlit. All heavy computational graph discovery steps have been pre-computed and stored within the local ./results directory. The dashboard serves real-time interactive physics networks, dynamic in silico knockout math, and subpopulation comparative queries without latency.

```
$ cd "D:\new project on 10102026"
# Navigate to the project root workspace
$ streamlit run dashboard/app.py
# Launch dashboard locally (opens default browser at http://localhost:8501)
$ streamlit run dashboard/app.py --server.port 8502
# Optional: specify alternate port if 8501 is occupied
$ streamlit run dashboard/app.py --server.headless true
# Optional: execute in headless mode for remote cloud server hosting
```

PRO-TIP: VERIFYING ARTIFACT AVAILABILITY & DIRECTORY PATH

Upon launch, the sidebar checks the ./results folder for key assets: top_master_regulators.csv, stable_causal_edges.csv, in_silico_knockout_effects.csv, subpopulation_causal_diff.csv, and drug_target_interactions.csv. If you store outputs in an alternate location, update the 'Artifacts Directory' path directly in the sidebar.

2. Module Walkthrough: Primary Causal Network & In Silico Perturbations

Tab 1: Causal Network Explorer

The Causal Network Explorer visualizes the entire 55-node, 69-edge directed causal network using interactive PyVis physics simulation. Unlike undirected correlation networks, directed arrows indicate the inferred direction of regulatory transmission.

- **Causal Out-Degree Filter:** Sliders allow you to isolate hub master regulators controlling 3, 4, or 5+ downstream targets.
- **Stability Frequency Cutoff:** Constrain the visible graph to edges meeting stringent bootstrap retention (default $\ge 60\%$, up to 100%).
- **Gene Search & Highlighting:** Typing a gene symbol (e.g., NPM1 or S100A9) highlights its direct causal parents (upstream controllers in gold) and direct children (downstream targets in cyan).
- **Physics Interaction:** Nodes can be clicked and dragged to untangle dense clusters; click 'Freeze Physics' in PyVis to fix positions for export.

Tab 2: Master Regulator Inspector

This module delivers deep-dive analytical profiling for individual driver genes. Selecting a candidate gene from the dropdown provides:

- **Topology Card:** Summarizes Master Regulator Score (MRS), global rank, out-degree, in-degree, and betweenness centrality.
- **Upstream vs. Downstream Projections:** Displays all immediate upstream regulators and all downstream reachable cascade targets.
- **Curated Oncogenic Biology Card:** Displays peer-reviewed functional mechanisms for established AML regulators (e.g., *NPM1* nucleolar stress, *FLT3* tyrosine kinase signaling, *BCL2* apoptotic threshold, *RUNX1* myeloid differentiation blockade).
- **Actionable Therapeutics Card:** Directly lists approved and investigational compounds mapped to the regulator.

Tab 3: In Silico Knockout Simulator (Pearl's do-Operator)

A marquee innovation of this platform is the implementation of Judea Pearl's structural causal intervention calculus ($do(X = 0)$). Rather than computing passive conditional correlations, the simulator perturbs the upstream node to zero and propagates the causal shock downstream through linear Structural Causal Model (SCM) path coefficients ($\beta_{u \to v}$) following forward topological sort order.

- **Knockout Selector:** Interactively simulate complete ablations of top master regulators: NPM1, NAMPT, S100A9, RNASE2, and FCN1.
- **Knockout Efficiency Slider (10% to 100%):** Model dose-dependent pharmacologic inhibition (e.g., 50% target knockdown vs. 100% CRISPR knockout).
- **Cascade Hierarchy Filter:** Inspect direct children (causal distance = 1) versus deep indirect cascade targets (distance ≥ 2).
- **Interactive Waterfall Bar Chart:** Displays percentage expression shifts ($\Delta \%$) across downstream genes, identifying catastrophic downstream collapses.

LANDMARK IN SILICO DISCOVERIES: TARGETED CASCADE COLLAPSES

1. **NPM1 -> MRPL42 (-62.5%):** Ablating NPM1 starves mitochondrial ribosomal protein translation, explaining why NPM1 disruption selectively impairs oxidative phosphorylation in leukemic stem cells.
2. **S100A9 -> THBS1 (-87.9%) & VCAN (-52.0%):** Silencing S100A9 shuts down downstream extracellular matrix protection and cell adhesion.
3. **RNASE2 -> S100A8/A9 (-32.1% / -25.4%):** RNASE2 acts as an apex trigger destabilizing the calprotectin alarmin complex.

2. Module Walkthrough: Differentiation Hierarchy & Translational Pharmacology

Tab 4: Leukemic Hierarchy & Subpopulation Rewiring

Therapeutic failure in AML often stems from Leukemic Stem Cells (LSCs) that harbor distinct transcriptional wiring compared to mature blasts. This module stratifies single cells into **Immature / Stem-like Blasts** (LSC-like / HSC-like / Prog-like, $n=535$) and **Differentiated Blasts** (Mono-like / Gran-like / cDC-like, $n=535$) and evaluates causal edge rewiring via 100x bootstrap stability.

• **The Four Functional Rewiring Categories:**

1. *Stemness-Maintained ($n=16$):* Active selectively in LSC-like blasts (e.g., HDC -> MS4A2 $\Delta \beta = +1.07$, PRG2 -> CLC $\Delta \beta = +0.85$, RNASE2 -> SRGN $\Delta \beta = +0.42$). These represent selective anti-leukemic stem cell vulnerabilities.
2. *Maturation-Induced ($n=26$):* Regulatory cascades emerging exclusively upon differentiation (e.g., HDC -> CXCL8 $\Delta \beta = -1.08$, TPSB2 -> MS4A2 $\Delta \beta = -0.81$, MTMR7 -> NAMPT $\Delta \beta = -0.48$).
3. *Conserved Core ($n=11$):* Baseline transcriptional machinery operating invariantly across all maturation states (e.g., SRGN -> ELANE, NAMPT -> RGS2).
4. *Context-Dependent ($n=16$):* Transitional or low-magnitude interactions.

Tab 5: Centrality & Target Ranking

Ranks all 55 network genes by graph-theoretical centrality metrics. Genes with high **Out-Degree** govern broad downstream transcriptional programs, while high **Betweenness Centrality** indicates vital network information bottlenecks. The composite Master Regulator Score (MRS) combines Out-Degree (50%), Betweenness Centrality (30%), and Downstream Reachability (20%).

Tab 6: Biological Pathway Enrichment

Displays overrepresentation analysis across Gene Ontology (GO Biological Process 2023) and KEGG Pathways. Interactive sliders allow filtering by adjusted p-value ($-\log_{10} p_{adj}$). Top enriched biological processes include **Innate Immune Defense** ($p = 3.34 \times 10^{-4}$), **Positive Regulation of Apoptotic Signaling** ($p = 1.98 \times 10^{-3}$), and **NF-κB Transcription Factor Activation** ($p = 9.13 \times 10^{-3}$).

Tab 7: Targeted Therapeutics & Drug Repurposing

Bridges causal discovery with clinical pharmacology. Cross-references the top 10 master regulators against ChEMBL and DGIdb, curating **31 actionable therapeutic candidates** (12 FDA-Approved, 19 Investigational).

• **Lead Therapeutic Candidates in AML:**

- **NAMPT:** Daporinad (FK866) & Teglarinad (KPT-9274) — Triggers acute NAD+ exhaustion, starving metabolically addicted leukemic blasts.
- **NPM1:** Actinomycin D & Selinexor (KPT-330) — Restores p53 tumor suppression via selective cytoplasmic NPM1 degradation and XPO1 inhibition.
- **S100A9/A8:** Tasquinimod & Auranofin — Blocks TLR4/RAGE alarmin signaling, clearing bone marrow niche chemoresistance.
- **TPSAB1:** Midostaurin (FDA approved) — Multi-kinase inhibitor targeting blast degranulation and survival.

Tab 8: Preprocessing & QC Audit

Provides complete transparency into Scanpy single-cell quality control metrics: cell filtering (>50 genes), gene filtering (>5 cells), mitochondrial exclusion (<25% MT reads), library normalization (10^4 target), and Seurat highly variable gene dispersion selection (top 80 HVGs).

3. Data Export & Downstream Bioinformatics Integration

All analytical outputs generated by the platform are exportable in standardized bioinformatics formats for immediate downstream analysis in R (Seurat, Bioconductor) or Python (Scanpy, NetworkX):

Artifact File	File Format	Primary Application & Downstream Utility
top_master_regulators.csv	CSV Table	Ranked driver leaderboard; input for CRISPR/shRNA screens and survival analysis.
stable_causal_edges.csv	CSV Table	Stability-selected causal links with bootstrap frequencies ($\ge 60\%$).
causal_graph.graphml	GraphML (XML)	Cytoscape & Gephi compatible DAG file with node and edge attributes.
in_silico_knockout_effects.csv	CSV Table	Pearl do-operator simulated expression shifts ($\Delta \%$, z-score, causal distance).
subpopulation_causal_diff.csv	CSV Table	Hierarchy rewiring metrics ($\Delta \beta = \beta_{imm} - \beta_{diff}$) for LSC targeting.
drug_target_interactions.csv	CSV Table	Actionable compounds curated from ChEMBL/DGIdb with clinical trial status.
results/*.png	300 DPI PNG	Publication-grade vector-quality figures formatted for manuscripts and grants.

4. Troubleshooting & Frequently Asked Questions (FAQ)

Q1: Port 8501 is already in use by another service.

Streamlit defaults to port 8501. Launch with streamlit run dashboard/app.py --server.port 8502 or terminate existing background processes using PowerShell: Get-Process -Name streamlit | Stop-Process.

Q2: Dashboard displays 'Awaiting Artifacts' or warnings in sidebar.

Ensure your working directory is the repository root (D:\new project on 10102026). Verify that ./results contains the exported CSV files. You can enter an absolute path directly in the sidebar's 'Artifacts Directory' text input.

Q3: Force-directed network graph renders slowly or lags during dragging.

Ensure hardware acceleration is enabled in your web browser settings (Chrome/Edge: 'Settings > System > Use graphics acceleration when available'). Additionally, increase the 'Minimum Causal Out-Degree' slider to filter out peripheral nodes.

Q4: Can I run this dashboard on a remote cloud server or Docker container?

Yes. Launch with streamlit run dashboard/app.py --server.headless true --server.address 0.0.0.0 --server.port 8501 to allow external reverse proxy or browser connections.

TECHNICAL SUPPORT & CITATIONS

If you encounter pipeline issues or require custom algorithmic extensions, consult the repository README at README.md. Please cite van Galen et al. (Cell 2019) and the Py-Why causal-learn consortium.